

Nilesh Parmar

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years** across **payments**, **GPU validation**, and **AI infrastructure**. Built a **RabbitMQ**-based payment retry system at Paras that recovered **30%** of failed transactions, and debugged GPU driver memory leaks at Nvidia using **Valgrind**. Currently building **LLM**-powered features at IpserLab using **Next.js**, **Gemini**, and **RAG** pipelines.

SKILLS

Languages: Java, Python, JavaScript/TypeScript, Go, SQL

Backend & Frontend: Spring Boot, Node.js, FastAPI, REST, GraphQL, gRPC, WebSocket; React, Next.js

Distributed Systems: Kafka, RabbitMQ, Redis, Microservices, Event-Driven Architecture

Cloud & DevOps: AWS (EKS, EC2, S3, CloudWatch, CloudFront), Docker, Kubernetes, GitHub Actions, Prometheus

Databases: PostgreSQL, MySQL, MongoDB, Redis, Qdrant

AI/LLM Tooling: LangChain, Google Gemini, Ollama, RAG Pipelines, Prompt Engineering; **ML:** XGBoost, Whisper

WORK EXPERIENCE

Software Engineer | IpserLab, USA

Oct 2025 – Present

- Built an **AI Quiz Generator** feature using **Next.js** and **Google Gemini 2.5** API, enabling admins to dynamically generate complete quizzes from inputs including topic, goal, tone, and question count, powering the AI content engine for a lead generation platform processing **5K+ quizzes/month** and lifting lead conversion by **20%**
- Engineered **structured JSON schema** prompting with **Gemini's** response schema config to enforce consistent, parseable quiz output, and implemented a **RAG pipeline** to ground generated questions in domain-specific knowledge for improved accuracy, relevance, and consistent content quality across diverse quiz topics
- Secured a server-side API route in **Next.js App Router** with **prompt engineering** and structured response handling, keeping credentials backend-protected while delivering real-time AI-generated titles, descriptions, and multiple-choice questions

Software Engineer | Paras Industries, India

Sep 2021 – Aug 2023

- Owned **backend services** and **REST APIs** for a high-traffic **B2C e-commerce** platform supporting **2K+ daily users** and reliable, end-to-end transactional payment workflows across multiple regions and product categories
- Built a **distributed retry system** with **RabbitMQ** and in-memory caching, improving payment reliability by **30%**, recovering **\$2K+ daily revenue** from failed transactions and ensuring transactional consistency at scale
- Revamped the frontend with **React** and **React Router**, implementing code-splitting and **CloudFront** CDN, reducing page load times by **50%** and improving checkout conversion rates across the product catalog
- Deployed **microservices** on **AWS EKS** with auto-scaling and cache pre-warming, and built a **CloudWatch** observability platform that cut incident resolution time by **60%** during traffic spikes

Software Engineer | Nvidia, India

Apr 2021 – Aug 2021

- Authored a **GPU driver validation framework** with **150+ Python** unit tests to increase test coverage from **63% to 77%**, reducing production bugs by **25%** and validating driver functionality across **10+ GPU architectures**
- Diagnosed and fixed **3 critical memory leaks** using **Valgrind** and custom profiling to eliminate daily crashes, resolving stability issues for **20+ developers** and reducing memory-related incidents through improved debugging practices
- Resolved **3 complex race conditions** in GPU driver code to enhance system stability, preventing intermittent failures in customer deployments and reducing production incidents by improving thread-safety mechanisms

Software Engineer | Nimbeshwar Technologies, India

Apr 2020 – Mar 2021

- Designed and launched a core **Payment API** using **Java (Spring Boot)** and **PostgreSQL**, improving transaction reliability by **20%**, reducing response times by **4x** and processing **10K+ daily transactions** at scale
- Implemented **Redis caching** with **TTL-based invalidation** and optimized database queries to boost system throughput, **reducing database load by 40%** under high-traffic, concurrent-load conditions across regional deployments
- Automated **CI/CD pipelines** with **GitHub Actions** and **Docker**, cutting deployment time from **3 hours to 45 minutes** and accelerating release velocity for the engineering team

PROJECTS

Academic Research Assistant Chatbot | GitHub

- Engineered a distributed RAG-based research assistant using Akka Cluster, enabling parallel PDF processing, horizontal scalability, and low-latency (< 2s) query responses
- Built an actor-based, event-driven pipeline integrating the Qdrant vector database and Ollama LLMs to deliver semantic search with citation-backed responses via REST APIs and UI

Therapist Copilot | GitHub

- Designed a real-time AI backend using FastAPI and WebSocket streaming for low-latency transcription via Deepgram/Whisper, processing audio with sub-second response times
- Integrated Google Gemini LLMs via LangChain for real-time risk assessment (suicide/self-harm detection), deploying a containerized, production-ready service with REST APIs and monitoring

Real-Time Fraud Detection Engine | GitHub

- Architected an end-to-end, event-driven fraud detection pipeline using Kafka, Go, and Redis, processing 50+ TPS at sub-100ms latency with dead-letter queue recovery, retry logic, and Prometheus monitoring for zero silent failures
- Built a production-grade ML scoring service (FastAPI + XGBoost, ROC-AUC 0.976, PR-AUC 0.902) with 22 real-time features, SHAP explainability, and feature drift detection deployed via Docker Compose with a WebSocket React dashboard

EDUCATION

Northeastern University | MS in Information Systems

Sep 2023 – Aug 2025

University of Mumbai | BE in Electronics and Telecommunication

Sep 2015 – Nov 2020